Agentic AI for Robust and Socially-Aware Autonomous Driving

Proactive Social Compliance Modeling using Agentic Theory of Mind

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Project Proposal Report

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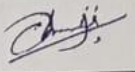
Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Autonomous vehicles (AVs) still struggle to understand pedestrian intentions before a crossing occurs. Most existing systems rely on trajectory prediction or binary crossing classification, making them reactive rather than proactive. Such approaches often fail to capture subtle behavioral cues such as hesitation, distraction, or sudden crossing intent. In addition, many existing models are trained on Western datasets and operate as black-box systems, limiting their adaptability to different traffic cultures and reducing trust in autonomous driving systems.

This research proposes a Theory-of-Mind-inspired agentic AI system for predicting pedestrian mental states and generating proactive driving decisions. The core component is a two-layer Long Short-Term Memory (LSTM) network that models temporal dependencies in pedestrian movements. Video frames from the PIE dataset are processed using OpenCV and MediaPipe to extract 36 motions, pose, behavioral, and contextual features, which are organized into 30-frame sequences representing one second of behavior. The LSTM classifies pedestrians into six mental states: Waiting, Hesitant, Committed, Distracted, Aggressive/Abnormal Gait, and Jaywalk.

A rule-based agent maps these predicted states to five proactive vehicle actions: maintain speed, slow down, brake, stop, or evasive maneuver. The system also includes a cultural adaptation module that adjusts decision thresholds using Sri Lankan pedestrian behavior statistics. Validation is conducted using the PIE dataset and CARLA simulation scenarios.

The expected outcome is an explainable and near real-time pedestrian behavior prediction framework that improves AV safety, supports safer urban mobility, and increases public trust in autonomous vehicle technologies.

Keywords: Long Short-Term Memory (LSTM), pedestrian intention prediction, autonomous vehicles, explainable AI, Theory of Mind

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TABLE OF CONTENT

Contents

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

TABLE OF CONTENT 5

LIST OF TABLES 6

LIST OF FIGURES 7

LIST OF ABBREVIATIONS 8

1. INTRODUCTION 9

2. OBJECTIVE 15

3. METHODOLOGY 16

4. HIGH-LEVEL SYSTEM ARCHITECTURE 22

5. USER REQUIREMENTS 26

6. COMMERCIALIZATION PLAN 28

7. BUDGET AND JUSTIFICATION 29

8. WORK BREAKDOWN STRUCTURE (WBS) 30

9. GANTT CHART 31

REFERENCES 32

LIST OF TABLES

Table 1:Research Gap 14
Table 2:Mapping of Design Science Research Process to the Proposed AV System..... 16
Table 3:Proposed Features 17
Table 4:Mental States Mapping 18
Table 5:LSTM Architecture 19
Table 6:How AV take action according to mental states 20
Table 7:Cultural Adaptation 20
Table 8:Tools and Platforms 21
Table 9:System Processing Stages 22
Table 10:Scalability 26
Table 11:Security and Safety 26
Table 12:Functional Requirement 27
Table 13:Non-Functional Requirement 27
Table 14:Cost Estimation 29
Table 15:Budget Justification 29
Table 16:Work Breakdown Structure 30

LIST OF FIGURES

Figure 1:Overall System Diagram 23
Figure 2:Individual Component System Diagram 24
Figure 3:Data Flow 25
Figure 4:Gantt Chart 31
Figure 4:Gantt Chart 31

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AV	Autonomous Vehicle
LSTM	Long Short-Term Memory
ToM	Theory of Mind
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
PIE	Pedestrian Intention Estimation (dataset)
RNN	Recurrent Neural Network
GNN	Graph Neural Network
OpenCV	Open Source Computer Vision Library
CARLA	Car Learning to Act (open urban driving simulator)
DSRM	Design Science Research Methodology
fps	Frames Per Second
AIMS	Autonomous Intelligent Machines and Systems
IoT	Internet of Things
WBS	Work Breakdown Structure
JSON	JavaScript Object Notation
GPU	Graphics Processing Unit
WHO	World Health Organization

1. INTRODUCTION

1.1 Background and Context

Current autonomous vehicles (AVs) are largely reactive, responding to pedestrian behavior only after it occurs. They rely on trajectory prediction models to estimate movement from past positions but cannot determine whether a pedestrian is hesitant, distracted, likely to jaywalk, or displaying abnormal gait. This limitation makes AVs reactive rather than proactive, waiting for pedestrians to act instead of anticipating intentions, which can lead to injuries, liability for manufacturers, and reduced public trust.

Pedestrians are directly affected, making up 23% of global road fatalities [5]. AV manufacturers face challenges in deploying vehicles safely in diverse environments, and in Sri Lanka, pedestrians face high risks at unsignalized crossings [3][4]. Globally, 1.19 million people died in road accidents in 2021, with pedestrian deaths rising 12.9% from 2013 to 2016 [5]. Local studies show 61% signal compliance, 7.45% distracted walking at 1.08 m/s, and higher-risk behaviors among males and elderly [3][4]. Datasets like PIE indicate pedestrian hesitation levels from 0.33 to 0.66 [1], highlighting the need to understand not just where pedestrians move but why.

Theory of Mind (ToM) is the ability to infer others' beliefs, intentions, and desires to predict their actions. In autonomous driving, this helps vehicles anticipate pedestrian behavior, such as hesitation or crossing intentions. Computationally, ToM can be implemented by inferring hidden pedestrian states from observable cues such as motion and trajectory. For example, Chandra et al. [10] introduce StylePredict, a Machine Theory of Mind framework that infers latent driving behaviors from trajectory data, demonstrating how ToM reasoning can be applied to interpret agent intentions. This approach can be adapted to pedestrian intention prediction by reasoning about mental states like hesitation or commitment before a crossing occurs, improving proactive decision-making in AV systems.

Agentic AI extends AV capabilities by perceiving micro-movements, reasoning via LSTM networks, planning pedestrian responses, selecting vehicle actions (maintain, slow down, brake, stop, evasive), and generating human-readable explanations. A rule-based agent converts inferred states into transparent decisions, providing human-auditable rules, safety fallbacks, and threshold adjustments for local norms without retraining [15][17].

Pedestrians show micro-movements before crossing, such as head turns, small steps, weight shifts, and gaze direction. Detecting these cues allows AVs to respond proactively, with 1.5 seconds of early detection enabling safe deceleration. Behavior is influenced by traffic culture, which varies across countries. Western-trained AVs may misinterpret behaviors in Sri Lanka, where 61% comply with signals, 7.45% walk distractedly, and risky behaviors are higher among males and elderly [3][4]. Threshold-based cultural adaptation modifies decision boundaries using local data without retraining.

Existing AV approaches including trajectory prediction, binary crossing prediction, Western-trained models, black-box neural networks, and sequential LSTM/RNN models [1][2][5][6][14][15] are limited, reactive, lacking intention understanding, failing in

developing world contexts, non-explainable, or prone to error accumulation. Even advanced deep learning methods incorporating social interactions (e.g., Zhang et al., 2024) struggle to detect subtle pre-crossing behaviors. The proposed system integrates five components: Theory of Mind, Agentic AI, rule-based decision-making, social compliance adaptation, and explainable AI. Validation uses CARLA simulations to demonstrate proactive and culturally aware AV behavior.

This project aligns with SDG 3.6, aiming to halve global road deaths, and SDG 11, contributing to safer, intelligent urban mobility[21]. At SLIIT, it belongs to the AIMS (Autonomous Intelligent Machines and Systems) cluster, focusing on explainable AI, cognitive modeling, computer vision, deep learning, and culturally adaptive autonomous systems.

1.2 Literature Review

Previous research on pedestrian intention prediction for autonomous vehicles (AVs) was reviewed to understand the current approaches used in this field. Existing studies mainly focus on different methodological approaches such as trajectory-based prediction, pose-based behavior analysis, temporal deep learning models, gait-based behavioral analysis, culture-aware pedestrian modeling, and explainable AI approaches for AV decision-making. This review evaluates these approaches, discusses their contributions, and identifies the limitations that motivate the proposed research.

EfficientPIE (Qu et al., 2025) [18]
EfficientPIE proposes a lightweight transformer model designed for real-time pedestrian crossing intention prediction using single-frame observations from the PIE dataset. The model demonstrates a 7.4× faster inference speed while maintaining high prediction accuracy, showing promise for real-time deployment in autonomous vehicles. Despite these advantages, the approach has several

limitations: it lacks explainability, relies exclusively on Western datasets, performs only binary classification (crossing vs. not crossing), and captures limited temporal context. This study highlights the trade-off between inference speed and interpretability, as well as the challenge of modeling more nuanced pedestrian behaviors

Zhang & Berger Review (2023) [6]
This paper presents a systematic taxonomy of deep learning methods for pedestrian behavior prediction, including LSTMs, RNNs, transformers, and graph neural networks (GNNs). It evaluates their performance, input features, and metrics across multiple urban datasets, providing a clear overview of the current state-of-the-art approaches.

Limitations:The review focuses mainly on technical analysis without proposing new predictive methods. It does not offer solutions for cross-cultural adaptation or real-time inference, highlighting the ongoing need for models that are both accurate and explainable, with transferable reasoning suitable for diverse contexts.

Cross-Cultural Model Transferability (Zhang et al., 2024) [7]
This study investigates how pedestrian prediction models trained in one country perform in another, using high-fidelity simulator data from Germany and Japan. The results reveal a 15-20% performance drop when models are applied across cultures, despite identical scenarios and simulation parameters.

Limitations: While it documents the transferability gap, it proposes no adaptation mechanism and is limited to Germany-Japan contexts. The study demonstrates that cultural adaptation is essential for accurate pedestrian intention prediction.

Spatial-Temporal-Spectral LSTM (Zhang et al., 2023) [8]
This paper proposes an LSTM model that incorporates spatial, temporal, and spectral features for pedestrian trajectory prediction. By using discrete Fourier transforms, it captures motion patterns like walking rhythm and hesitation cycles, improving transferability across datasets.

Limitations: The model is computationally heavy, remains a black-box, and focuses solely on trajectory rather than intention or mental state prediction. It shows that multi-modal feature extraction improves accuracy but does not address explainability or proactive intention detection.

Driver Temporal Segmentation (Zhang et al., 2025) [9]
This paper introduces a 4-stage Theory of Mind (ToM) framework to temporally segment pedestrian crossing intentions, modeling the negotiation process between drivers and pedestrians at unsignalized crossings. The stages include coarse-level intention estimation, fine-level action prediction, boundary detection, and memory updates, enabling the model to capture when intentions change.

Limitations**:** The model does not classify actual pedestrian mental states (e.g., hesitant, distracted, committed). it detects when intentions change, but not what the mental state actually is. It remains a black-box, lacks cultural adaptation, and does not consider real-time deployment. While it establishes a strong theoretical foundation for ToM-based pedestrian modeling, it does not provide practical solutions for explainable, multi-state, real-world prediction.

GTransPDM (Chen et al., 2024) [11]
This paper introduces a graph-embedded transformer model for pedestrian crossing intention prediction. It combines graph neural networks to model interactions among pedestrians and vehicles with transformers for capturing temporal dynamics, achieving 92% accuracy on the PIE dataset. The model is proactive in predicting intentions by considering social interactions and temporal context.

Limitations: The model is computationally heavy, functions as a black-box, and relies on Western-only datasets. It uses binary classification, missing nuanced pedestrian mental states (e.g., hesitant, distracted, committed), so while it predicts if an intention exists, it does not classify the actual mental state. Cultural adaptation and real-time feasibility are also not addressed.

PedAST-GCN (Liu et al., 2024) [12]
This study proposes a spatial-temporal attention graph convolution network that weights social interactions among road users to predict pedestrian intentions. Attention mechanisms highlight influential neighbors, improving predictive efficiency.

Limitations: High computational cost, low interpretability beyond attention weights, no cultural adaptation, and binary classification only. It reinforces the need for models that balance accuracy, efficiency, and explainability.

MonoPIC -Radhakrishna et al. (2024) [13]
Radhakrishna et al. introduced MonoPIC, a monocular pedestrian intention classification framework designed for real-time IoT edge deployment using pose features derived from MediaPipe [4][13]. The system extracts quaternion-based pose features from monocular video frames and applies ID3 decision tree models to classify pedestrian intention states. Their approach achieved an average testing accuracy of 83.56% with approximately 48 milliseconds latency, demonstrating that pose-based features can enable real-time pedestrian intent classification on edge devices.

Although MonoPIC shows promising real-time performance, its reliance on decision-tree-based classification limits its ability to model temporal dependencies across multiple frames, which is an important aspect for predicting evolving pedestrian behaviors

Pedestrian Vision Language Model for Intentions Prediction (PedVLM)[14]
Zhang et al. [14] proposed PedVLM, a Vision-Language Model designed for pedestrian intention prediction with an emphasis on explainable artificial intelligence (XAI). The objective of the study is to improve autonomous vehicle decision-making by enabling models to not only predict pedestrian intentions but also provide interpretable explanations for those predictions. The methodology integrates visual perception with language-based reasoning, where visual features extracted from pedestrian scenes are combined with natural language representations describing pedestrian behavior and environmental context. By linking visual cues with textual explanations, the system can generate interpretable reasoning about why a pedestrian may cross or hesitate.

The study demonstrates that combining vision and language representations improves both prediction performance and interpretability compared to traditional vision-only approaches. This contributes to the development of explainable intention prediction systems, which are important for safety-critical applications such as autonomous driving.

The model requires large multimodal datasets containing both visual data and textual annotations, which increases data preparation complexity and computational cost. Additionally, the framework focuses primarily on crossing intention prediction and does not explicitly incorporate detailed pose-based behavioral features or cultural traffic variations

Intention-Aware Model Predictive Control (Yamamoto et al., 2025) [17]
This paper integrates predicted pedestrian intentions into a model predictive control framework for vehicle trajectory planning. By anticipating pedestrian actions, the system improves safety and efficiency in control decisions.

Limitations: Focuses on control rather than intention inference, lacks explainability, and provides no cross-cultural validation. It highlights the importance of intention estimation for proactive AV control but leaves inference and interpretability gaps.

Learning Pedestrian Actions to Ensure Safe Autonomous Driving (Huang et al., 2023)[19]

This paper proposes a novel multi-task Transformer encoder-decoder model for simultaneously predicting pedestrian actions and trajectories using only camera data from the ego vehicle. The authors also present a clear taxonomy for pedestrian behavior, addressing inconsistencies in how behaviors are defined in previous work. The model is evaluated on the JAAD dataset, CARLA simulation data, and real shuttle data from a university campus, achieving about 81% action prediction accuracy and outperforming a comparable LSTM model.

Limitations: The approach still uses a binary action/trajectory prediction setup, not a mental state taxonomy (e.g., hesitant, distracted, committed). It also requires evaluation across more diverse real-world scenarios beyond campus data, and it does not integrate explainability into the prediction outputs. While it shows that multi-task and Transformer architectures can improve short-term action prediction, it stops short of classifying detailed pedestrian intent states relevant for proactive AV decision-making

1.3 Research Gap

Gap Area	Current Limitation / Evidence	How This Research Addresses It
Mental State Taxonomy	Binary or action-based models only [1][6][11][12][18]; Driver Temporal Segmentation detects *when* intentions change, not *what* the state is [9]	Introduces 6-state taxonomy (Waiting, Hesitant, Committed, Distracted, Aggressive, Jaywalk) built on ToM framework
Cultural Adaptation	Western-only models; 15-20% cross-country performance drop [7]; Sri Lankan norms differ significantly [3][4]	Threshold-based adaptation for Sri Lankan pedestrian behavior without retraining
Explainable AI	Black-box models dominate [11][12][18]; experimental neuro-symbolic XAI only [15][14], not real-time or culturally aware	Implements practical, real-time neuro-symbolic XAI with auditable rule templates and cultural threshold adaptation
ToM Framework	Detects intention changes but does not classify mental states [9]	Extends ToM framework to classify actual pedestrian

		mental states using 6-state taxonomy
Real-World Validation	Simulator-only or limited campus data [9][18];	Validates using PIE dataset (real) + CARLA simulations for diverse scenarios
Real-Time Multi-State Trade-off	EfficientPIE [18] is fast but binary; GTransPDM [11]/PedAST-GCN [12] accurate but slow	Combines 36-feature real-time agent with multi-state inference and neuro-symbolic reasoning for proactive prediction

Table 1:Research Gap

Explanation of Gaps

Binary Classification Misses Safety States: Most models only predict crossing vs. not crossing. Critical safety states like hesitation or distraction are ignored, limiting proactive responses.

Action Prediction ≠ Mental State Inference: Predicting "will cross" does not indicate why. Six distinct states allow AVs to respond appropriately to hesitation, distraction, or abnormal gait.

Cultural Adaptation is Absent: Models trained on Western data mis predict pedestrian behavior in developing countries. Threshold-based adaptation adjusts to local norms without retraining.

Black-Box Decisions Undermine Trust: Explainability is required for passenger trust, certification, and post-incident analysis. Neuro-symbolic XAI provides auditable, interpretable reasoning.

ToM Frameworks Lack State Taxonomy: Existing Theory of Mind models detect when intentions change but do not classify the actual mental states. Adding a 6-state taxonomy fills this gap.

2. OBJECTIVE

2.1 Main Objective

To develop a Theory-of-Mind-inspired agentic AI system that predicts six pedestrian mental states (Waiting, Hesitant, Committed, Distracted, Aggressive/Abnormal Gait, Jaywalk) from 36 micro-movement features, generates explainable proactive driving actions, adapts to cultural traffic norms.

2.2 Specific Objective

1. SO1: Feature Extraction
   Extract 36 micro-movement, pose, motion, and contextual features from PIE dataset videos using OpenCV and MediaPipe, with preprocessing including frame-rate normalization and data augmentation, to ensure robust input for classification.

2. SO2: Mental State Classification
   Train a two-layer LSTM network on 30-frame sequences to classify six pedestrian mental states (Waiting, Hesitant, Committed, Distracted, Aggressive, Jaywalk), extending the ToM framework into practical multi-state prediction, with cross-validation to evaluate generalization.

3. SO3: Explainable Decision Agent
   Implement a neuro-symbolic rule-based agent mapping inferred mental states to five proactive vehicle actions (MAINTAIN, SLOW_DOWN, BRAKE, STOP, EVASIVE), generating human-readable reasoning templates auditable for safety and trust.

4. SO4: Cultural Adaptation
   Develop a threshold-based adaptation module to adjust decision boundaries based on pedestrian behavior data. enabling culturally-aware predictions.

5. SO5: Simulation and Real-Time Validation
   Integrate the complete system with CARLA simulator and create validation scenarios covering all six mental states to ensure real-time feasibility and safety.

6. SO6: Evaluation and Performance Metrics
   Evaluate classification accuracy, per-class F1-score, early detection time, explanation quality, cultural adaptation gain, and overall proactive safety performance to ensure practical applicability.

3. METHODOLOGY

3.1 Research Approach

This research follows the Design Science Research Methodology (DSRM), appropriate for developing and evaluating innovative IT artifacts. DSRM involves six iterative steps.

Step	Description	Application in This Research
1	Problem Identification	Reactive AV systems cannot infer pedestrian mental states [1][6]; explainability absent [15][14]
2	Solution Objectives	6-state mental taxonomy, proactive actions, explainable AI
3	Design and Development	36-feature LSTM + rule-based agent (Sections 3.3-3.5) [8][11]
4	Demonstration	PIE dataset classification [1]; CARLA simulation scenarios [2]
5	Evaluation	Accuracy (>85%), F1 (>80%), early detection (>1.5s) [9][18]
6	Communication	Thesis, publications, presentations

Table 2:Mapping of Design Science Research Process to the Proposed AV System

This structured methodology ensures rigorous development and evaluation of the proposed agentic AI system.

3.2 Overall Methodology Pipeline

Phases:

- **Data Preparation:** PIE dataset [1] (53 videos, 300K+ frames) split into 70% train, 15% validation, 15% test.
- **Feature Extraction:** OpenCV → MediaPipe → 36 features (Motion, Pose, Behavioral, Context) [8][12][13].
- **Sequence Creation:** 30 consecutive frames = 1-second sequence; input shape: (batch_size, 30, 36) [19].
- **Label Generation:** Six mental states derived from intention scores (labels only, not input features) [1].
- **LSTM Training:** Two-layer LSTM; categorical cross-entropy; Adam optimizer; early stopping [19].
- **Rule-Based Agent:** Maps mental states to 5 proactive actions with explainable templates [15].
- **Cultural Adaptation:** Mainly focus on Sri Lankan traffic culture. data used only for threshold calibration, [3][4][16].

- **Validation:** PIE test set metrics, CARLA simulations, early detection, and XAI evaluation [2][18].

3.3 Technical Methods and Algorithms

3.3.1 Feature Extraction

- **Frame Extraction (OpenCV):** 30 fps, preserving temporal resolution [8].
- **Pose Estimation (MediaPipe):** 33 landmarks (head, torso, arms, legs) [8][12][13].
- **36 Feature Calculation:**

Category	Features	Count
Motion	current_speed, avg_speed, max_speed, acceleration, deceleration, speed_variance, step_frequency, left_step_length, right_step_length, pause_between_steps	10
Pose	upper_body_angle, lower_body_angle, head_angle, head_turn_frequency, shoulder/hip angles, foot angles, forward/lateral lean, body orientation change	12
Behavioral	pause_duration, hesitation_cycles, total_hesitation_time, distance_to_curb, distance_change_rate, temporal_movement_probability, looks_left, looks_right, direction_changes	9
Context	traffic_light_state, vehicle_distance, crosswalk_presence, road_width, pedestrian_density	5

Table 3:Proposed Features

Temporal movement probability is computed only from observable motion patterns to prevent label leakage [1][6].

3.3.2 Label Generation

Six mental states derived from intention scores [1]:

Mental State	Labeling Rule	Primary Features
Waiting	intention <0.33 & action='standing'	speed≈0, time_at_location>2s
Hesitant	0.33≤intention≤0.66	head_turn_freq>2, step_len<0.3, pause>1s

Committed	intention>0.66 or cross='crossing'	forward_lean>15°, distance_change_rate<0
Distracted	look='not-looking' >2s & speed<1.08 m/s	head_angle<-30°, look_freq<0.5
Aggressive/Abnormal Gait	speed>1.5 or gait_variability>0.4	step_timing_irregular, lateral_sway
Jaywalk	traffic_light='red' & cross='crossing'	traffic_light_state, cross

Table 4:Mental States Mapping

3.3.3 Sequence Creation

- Sliding window with stride 1: 30 frames = 1-second sequence [19].
- Input shape for LSTM: (batch_size, 30, 36).

3.3.4 LSTM Architecture

The Long Short-Term Memory (LSTM) network models temporal dependencies across sequences of pedestrian micro-movements. Following the findings of Huang et al. [19] that LSTM outperforms Transformers for 25+ frame sequences, a two-layer LSTM architecture is implemented.

Layer	Configuration	Purpose
Input	(30 time steps × 36 features)	1-second sequence capturing hesitation cycles
LSTM 1	128 units, return_sequences=True	Capture long-range temporal dependencies [8][9]
Dropout 1	0.3	Prevent overfitting during training [20]
LSTM 2	64 units	Compress sequence representation [19]
Dropout 2	0.3	Regularization to improve generalization [20]
Dense 1	32 units, ReLU	Non-linear feature combination [8]

Layer	Configuration	Purpose
Output Dense	6 units, Softmax	Mental state probabilities for 6-state taxonomy [1][10]

Table 5:LSTM Architecture

Training Configuration:

- **Loss Function:** Categorical Cross-Entropy [6]
- **Optimizer:** Adam, learning_rate=0.001 [20]
- **Batch Size:** 32 [8]
- **Early Stopping:** Patience 10 epochs to prevent overfitting [19]
- **Validation Split:** 15% of training data

3.3.5 Rule-Based Agent with XAI

- Maps six mental states to five proactive actions (MAINTAIN, SLOW_DOWN, BRAKE, STOP, EVASIVE) [14][15].
- Explanations generated using templates filled with feature evidence.

Predicted Mental State	**Confidence Threshold**	**AV Action**	**Reasoning**
Waiting	>0.7	**A1: MAINTAIN**	Pedestrian at curb with stable posture and normal traffic checking behavior. No immediate crossing intention detected. [1][9]
Hesitant	>0.7	**A2: SLOW_DOWN**	Pedestrian repeatedly scanning traffic, taking small steps and pausing. Crossing may occur suddenly when a traffic gap appears. [9][18]
Committed	>0.7	**A3: BRAKE**	Pedestrian walking steadily toward the road with forward body lean, indicating a strong crossing intention. [1][8]
Distracted	>0.6	**A2: SLOW_DOWN**	Pedestrian looking down (e.g., phone usage), reduced traffic checking, and slower walking speed, indicating reduced awareness of surroundings. [3][12]

Aggressive / Abnormal Gait	>0.6	**A4: STOP**	Pedestrian moving at unusually high speed or with irregular gait and minimal traffic checking, representing unpredictable behavior. [8][11][12]
Jaywalk	>0.6	**A3: BRAKE**	Pedestrian crossing during a red traffic signal or outside designated crosswalk areas, requiring anticipatory braking. [1][4][16]
Low Confidence	<0.6	**A2: SLOW_DOWN**	Model prediction confidence below safety threshold; precautionary speed reduction applied. [9][18]
Emergency / Imminent Collision	Distance < 5 m AND vehicle speed > 30 km/h	**A5: EVASIVE**	Collision risk imminent; vehicle performs evasive maneuver if adjacent lane is free, otherwise applies maximum braking. [2][17]

Table 6:How AV take action according to mental states

3.3.6 Cultural Adaptation Module

- According to culture traffic norms are different so here mainly focus on Sri Lankan traffic culture. Sri Lankan pedestrian data [3][4] used only to adjust thresholds.

Parameter	Baseline	Sri Lankan	Applied To
Signal Compliance	0.90	0.61	Waiting confidence ×0.68
Distracted Speed	1.4 m/s	1.08 m/s	Detection calibration
Distracted Prevalence	5%	7.45%	Sensitivity calibration

Table 7:Cultural Adaptation

3.4 Tools and Platforms

Tool	Purpose	Justification
PIE Dataset	Primary training data	Only dataset with intention scores [1]
OpenCV	Frame extraction	Industry standard, Python API [8]
MediaPipe	Pose estimation	Real-time, 33 landmarks, research-validated [12][13]

PyTorch	Deep learning	GPU-accelerated, dynamic computation graphs [8][9]
LSTM	Temporal sequence modeling	Outperforms Transformers for 25+ frame sequences [19]
CARLA Simulator	Validation	Realistic urban scenarios [2]
NumPy/Pandas	Data processing	Efficient array operations
Scikit-learn	Evaluation	Standard metrics

Table 8:Tools and Platforms

3.5 Data Collection and Ethics

- PIE Dataset: 53 videos, 300K+ frames, 1,800+ pedestrians, publicly available [1].
- Ethics: No new human data collected; all data anonymized and stored securely [5].

3.6 Validation Strategy

- **PIE Test Set:** Accuracy >85%, F1 >80%, confusion matrix for 6 states [1][6].
- **Early Detection:** Hesitant >1.5s, Jaywalk >1.8s before crossing [9][18].
- **CARLA Simulator:** 50 scenarios covering all 6 mental states, collision rate <5%, action appropriateness >80% [2].
- **XAI Quality:** Explanation relevance >4/5, causal accuracy >0.7 [14][15].
- **Cultural Adaptation:** Threshold adjustments improve accuracy by ≥10% over baseline [3][4][16].
- **Ablation Study:** Incrementally adds LSTM, rules, XAI, and thresholds to demonstrate contribution.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed system is a neuro-symbolic agentic AI architecture designed to enable autonomous vehicles to anticipate pedestrian behavior before a crossing occurs. The system combines deep learning-based perception, symbolic reasoning, and culturally adaptive decision thresholds to generate proactive vehicle responses.

The architecture integrates three major capabilities:

7. **Data-Driven Neural Perception**
   A two-layer Long Short-Term Memory (LSTM) network analyzes temporal sequences of pedestrian micro-movements extracted from video frames to infer six pedestrian mental states. These states include Waiting, Hesitant, Committed, Distracted, Aggressive/Abnormal Gait, and Jaywalk. The model uses pose-based features extracted from the PIE Dataset (Pedestrian Intention Estimation).
8. **Symbolic Decision Reasoning**
   A rule-based agent converts predicted mental states into five proactive vehicle actions: Maintain Speed, Slow Down, Brake, Stop, and Evasive Maneuver. These rules provide interpretable reasoning paths, improving trust and auditability.
9. **Cultural Behavior Adaptation**
   Pedestrian behavior varies across traffic cultures. Here Sri Lankan pedestrian statistics are used to adjust decision thresholds without retraining the machine learning model, allowing the system to adapt to local traffic norms.

The overall architecture operates in **four main stages**, summarized in Table 4.1.

Stage	Component	Function
Stage 1	Feature Extraction Layer	Converts video frames into micro-movement features
Stage 2	LSTM Classification Layer	Infers six pedestrian mental states
Stage 3	Cultural Adaptation + Rule-Based Agent	Converts mental states into vehicle actions
Stage 4	Validation Layer	Evaluates system using dataset testing and simulation

Table 9:System Processing Stages

4.2 System Diagrams

Overall system Diagram

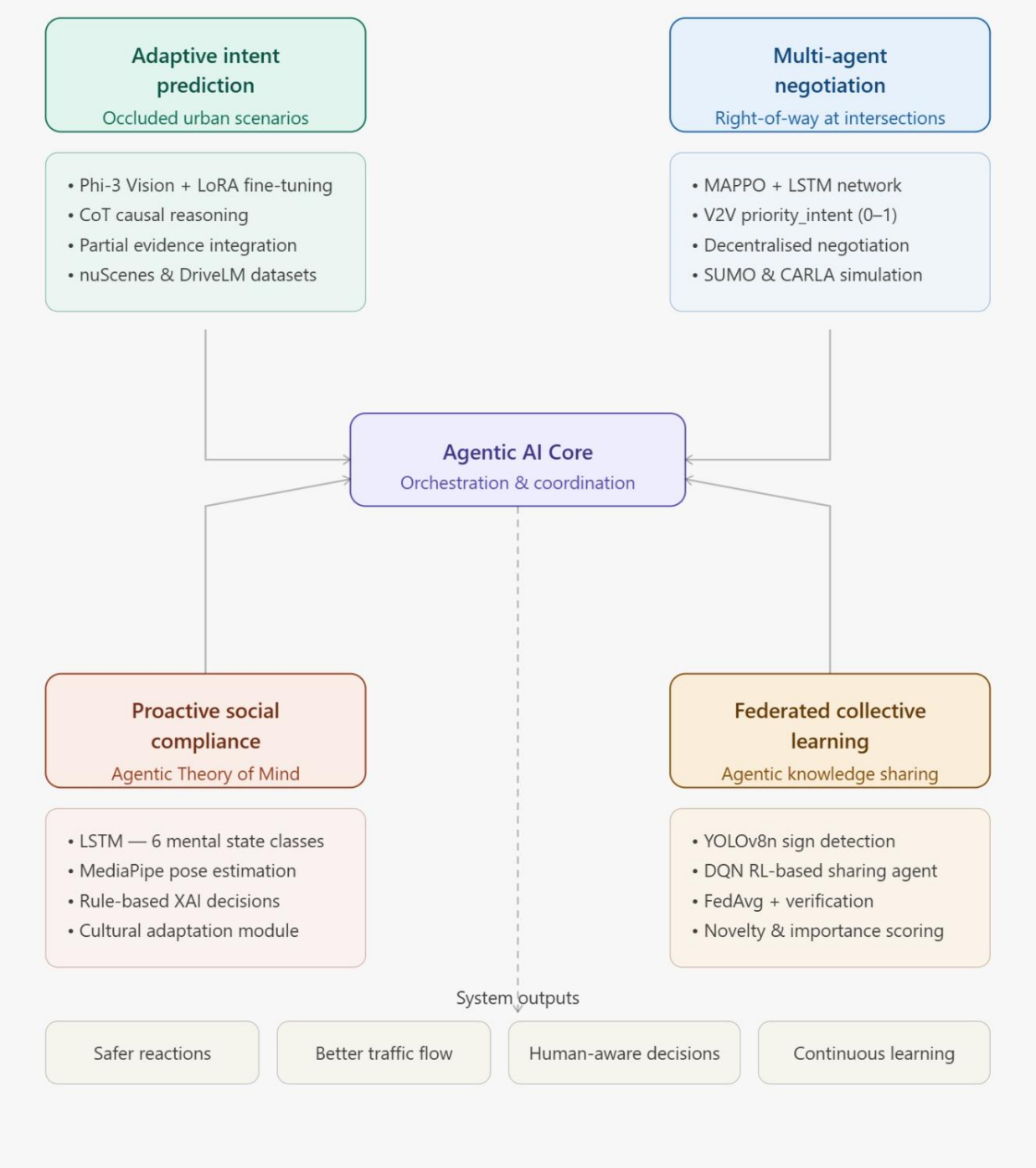


Figure 1:Overall System Diagram

Individual Component System Diagram

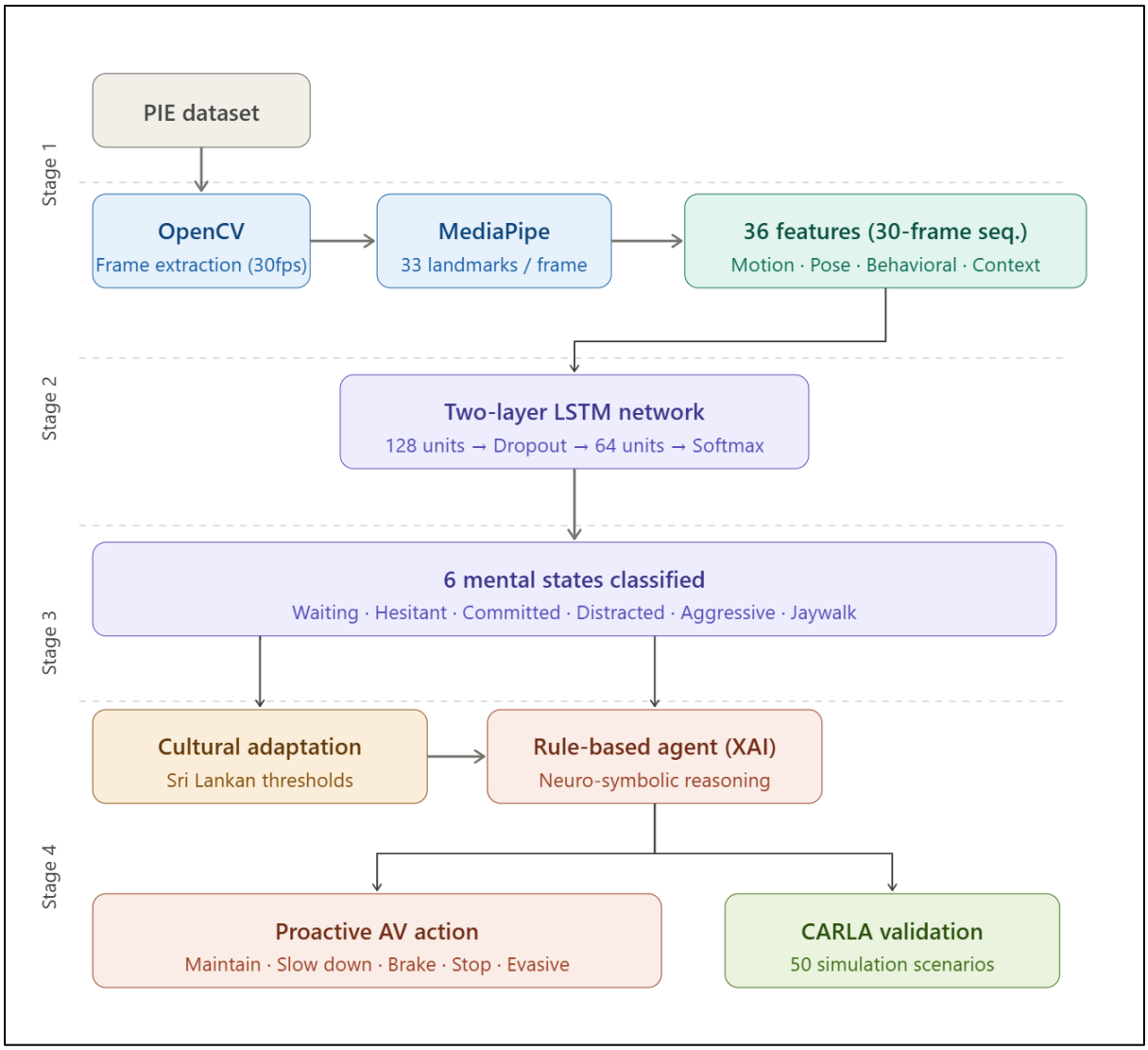


Figure 2:Individual Component System Diagram

4.3 System Components

The proposed system analyzes pedestrian behavior and recommends safe actions for autonomous vehicles. It consists of four main components.

Input Data: The system uses the PIE pedestrian dataset for training and testing.

Feature Extraction: Video frames are processed using computer vision techniques to detect body landmarks and extract motion, pose, behavioral, and contextual features.

Mental State Prediction:An LSTM neural network analyzes sequences of pedestrian movements and predicts six mental states: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk.

Decision Module:A rule-based agent converts predicted mental states into vehicle actions such as maintain speed, slow down, brake, or stop. If uncertainty or collision risk is detected, precautionary actions are applied.

4.4 Component Interactions

The components interact sequentially as follows:

- Pedestrian video data is processed into frames.
- Pose estimation extracts body landmarks.
- Micro-movement features are computed.
- Temporal sequences are fed into the LSTM network.
- The LSTM predicts pedestrian mental state probabilities.
- Cultural adaptation adjusts decision thresholds.
- The rule-based agent selects the safest vehicle action.
- The system outputs the action and explanation.

4.5 Data Flow

1. **Video Input:** Raw pedestrian videos from PIE dataset.
2. **Frame Extraction**: Frames extracted at 30 fps using OpenCV.
3. **Pose Estimation:** Detect 33 body landmarks per frame using MediaPipe.
4. **Feature Calculation:** Motion, pose, behavioral, and context features computed (36 features per frame).
5. **Sequence Generation:** Consecutive frames grouped into 30-frame sequences for temporal modeling.
6. **Mental State Prediction:** LSTM network predicts one of six pedestrian mental states.
7. **Decision Agent:** Rule-based module maps predicted state to AV action (maintain, slow down, brake, stop, or evasive) and generates a human-readable explanation.
8. **Cultural Adaptation:** Thresholds adjusted based on local pedestrian behavior before final action.

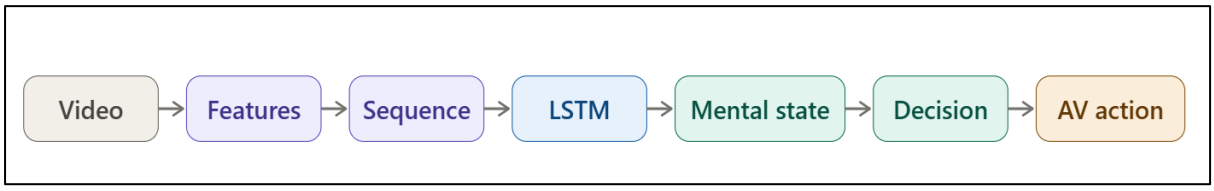


Figure 3:Data Flow

4.6 Scalability and Security Considerations

Scalability

Aspect	Design Strategy
Real-time processing	MediaPipe optimized for real-time pose estimation
Lightweight model	LSTM architecture with limited parameters
Cultural adaptation	Threshold calibration avoids retraining
Simulation testing	Scenarios can run in parallel simulations

Table 10:Scalability

Security and Safety

Risk	Mitigation
Low model confidence	Trigger safe slow-down action
Unexpected pedestrian behavior	Emergency braking or evasive maneuver
Data privacy	Use anonymized public datasets
Model failure	Rule-based fallback decision layer

Table 11:Security and Safety

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders for this system include autonomous vehicle (AV) manufacturers, who aim to integrate pedestrian intention prediction to enhance vehicle safety and improve decision-making. Traffic safety authorities are interested in reducing accidents and ensuring compliance with pedestrian safety regulations. Passengers and AV users benefit from safer driving and reduced risk of collisions. Pedestrians themselves are indirect stakeholders, as the system improves their safety by predicting mental states and enabling proactive AV responses. Finally, system developers and data scientists are responsible for implementing, maintaining, and refining the LSTM-based predictive models and rule-based agent to ensure accurate and reliable performance.

5.2 Functional Requirements

ID	Requirement	Priority
FR01	The system shall extract skeletal pose landmarks from pedestrian video frames using MediaPipe.	High
FR02	The system shall calculate all 36 features (motion, pose, behavioral, context) for each extracted frame.	High
FR03	The system shall create 30-frame (1-second) temporal sequences as input to the LSTM model.	High
FR04	The system shall classify pedestrian mental state into one of six classes with an associated confidence score.	High
FR05	The system shall map each classified mental state to one of five AV actions (MAINTAIN, SLOW_DOWN, BRAKE, STOP, EVASIVE).	High

FR06	The system shall generate a natural-language reasoning string explaining each AV action decision.	Medium
FR07	The system shall apply cultural threshold adjustments when a Sri Lankan context flag is enabled.	Medium
FR08	The system shall default to SLOW_DOWN when classifier confidence is below 0.60.	High
FR09	The system shall output classification results and action decisions with timestamp metadata.	Low
FR10	The system shall generate per-class evaluation reports including accuracy, precision, recall, and F1-score.	Medium

Table 12:Functional Requirement

5.3 Non-Functional Requirements

ID	Category	Requirement
NFR01	Performance	The system shall achieve >85% classification accuracy on PIE test set sequences.
NFR02	Latency	The end-to-end inference pipeline shall complete within 100ms per 1-second sequence.
NFR03	Reliability	The system shall maintain consistent classification across all six mental states with no class collapse.
NFR04	Explainability	Every AV action output shall be accompanied by a human-readable reasoning string.
NFR05	Adaptability	Cultural parameters (e.g., jaywalking thresholds) shall be configurable without model retraining.
NFR06	Privacy	The system shall process only skeletal landmark data, not raw facial or biometric information.
NFR07	Usability	The system interface shall provide clear, structured JSON-formatted output readable by AV middleware.

Table 13:Non-Functional Requirement

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

- Primary Market: Autonomous vehicle (AV) manufacturers and startups in Sri Lanka exploring smart mobility solutions.
- Secondary Market: Ride-hailing companies and logistic fleets adopting advanced pedestrian safety modules.
- Customer Persona:
    1. CTO or Safety Officer at AV startups/companies
    2. Concerned about pedestrian safety, regulatory compliance, and explainable AI for AV decisions
    3. Interested in cost-effective, scalable, locally adapted solutions

Value Proposition

- Proactive pedestrian intention prediction using a 6-state mental model.
- Real-time, culturally adapted decision-making according to traffic conditions.
- Explainable AI ensures trust, safety certification, and regulatory compliance.
- Reduced accident risk, liability, and enhanced public trust.

Revenue Model

A possible future commercialization pathway is licensing the pedestrian behavior prediction module to autonomous vehicle developers, mobility startups, or intelligent transportation solution providers after further validation and industry testing. The system could be integrated as a safety module in autonomous driving platforms[22].

Revenue could be generated through software integration and ongoing support services such as system updates, localization adjustments, and safety recalibration.

Indicative Pricing Model

- Integration Fee: approximately LKR 2,000,000 per vehicle for system integration and deployment.
- Annual Subscription: approximately LKR 50,000 per vehicle per year for software updates, threshold calibration, and technical support.

Cost Estimation

Resource	Estimated Cost (LKR)	Notes
Computing Hardware	250,000	GPU-enabled workstation with 8GB+8GB RAM, SSD, and peripherals
Software Licenses	50,000	Paid add-ons, simulation plugins, or cloud APIs if required

CARLA Simulator / Data Storage	25,000	Storage for videos, simulations, and model checkpoints
Human Effort	150,000	Undergraduate research & developer labor (3-4 months)
Total	**475,000**	Initial development cost

Table 14:Cost Estimation

Pricing Strategy

- Per Vehicle Integration: Recovers initial R&D cost, customized for AV companies.
- Subscription Model: Recurring revenue for updates, cultural adaptation, and support.

Competitive Advantage

- First Sri Lanka-focused, Theory-of-Mind-inspired AV pedestrian prediction system.
- Real-time, multi-state mental model vs. binary predictions in existing systems.
- Neuro-symbolic explainability supports auditability and trust
- Cultural adaptation without retraining for local traffic behaviors.

Intellectual Property Considerations

- Copyright protection on software code and AI model architecture.
- Patent application potential for the 6-state pedestrian mental model applied in AVs.
- Trade secrets maintained for threshold calibration methods and rule-based XAI templates.

7. BUDGET AND JUSTIFICATION

Item	Estimated Cost (LKR)	Justification
Hardware Costs	250,000	GPU workstation for model training, simulations, and testing
Software Licenses	50,000	Paid tools/plugins if needed (OpenCV, MediaPipe free, but any optional paid add-ons)
Cloud Services / Storage	10,000	Backup storage for dataset, model checkpoints, and simulation outputs
Human Effort	150,000	Research, preprocessing, model training, validation, documentation
Total	**460,000**	Covers development and simulation resources

Table 15:Budget Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS	Task	Deliverable	Hours	Week
1.1	Literature Review	20 papers reviewed	40	1-2
1.2	Gap Analysis	Gap statement, 6 research questions	15	2
1.3	Methodology Design	Complete methodology section	15	3
1.4	Architecture Design	Architecture + data flow diagrams	15	4
1.5	Proposal Writing	Complete proposal	15	4
2.1	Data Preparation	Processed PIE dataset	30	5
2.2	Feature Extraction	36 features extracted	60	6-7
2.3	Sequence Creation	30-frame sequences	20	8
2.4	Label Generation	6-state labels	20	8
2.5	LSTM Development	Trained LSTM model	60	9-10
2.6	Rule-Based Agent	Rule agent + XAI templates	30	11
2.7	Cultural Adaptation	Threshold calibration	15	11
2.8	CARLA Setup	50 validation scenarios	25	12
3.1	Evaluation	Complete results	40	13-14
3.2	Analysis	Findings and limitations	20	14
3.3	Thesis Writing	Complete thesis	60	15-16
3.4	Presentation Prep	PP2 presentation	20	16

Table 16:Work Breakdown Structure

9. GANTT CHART

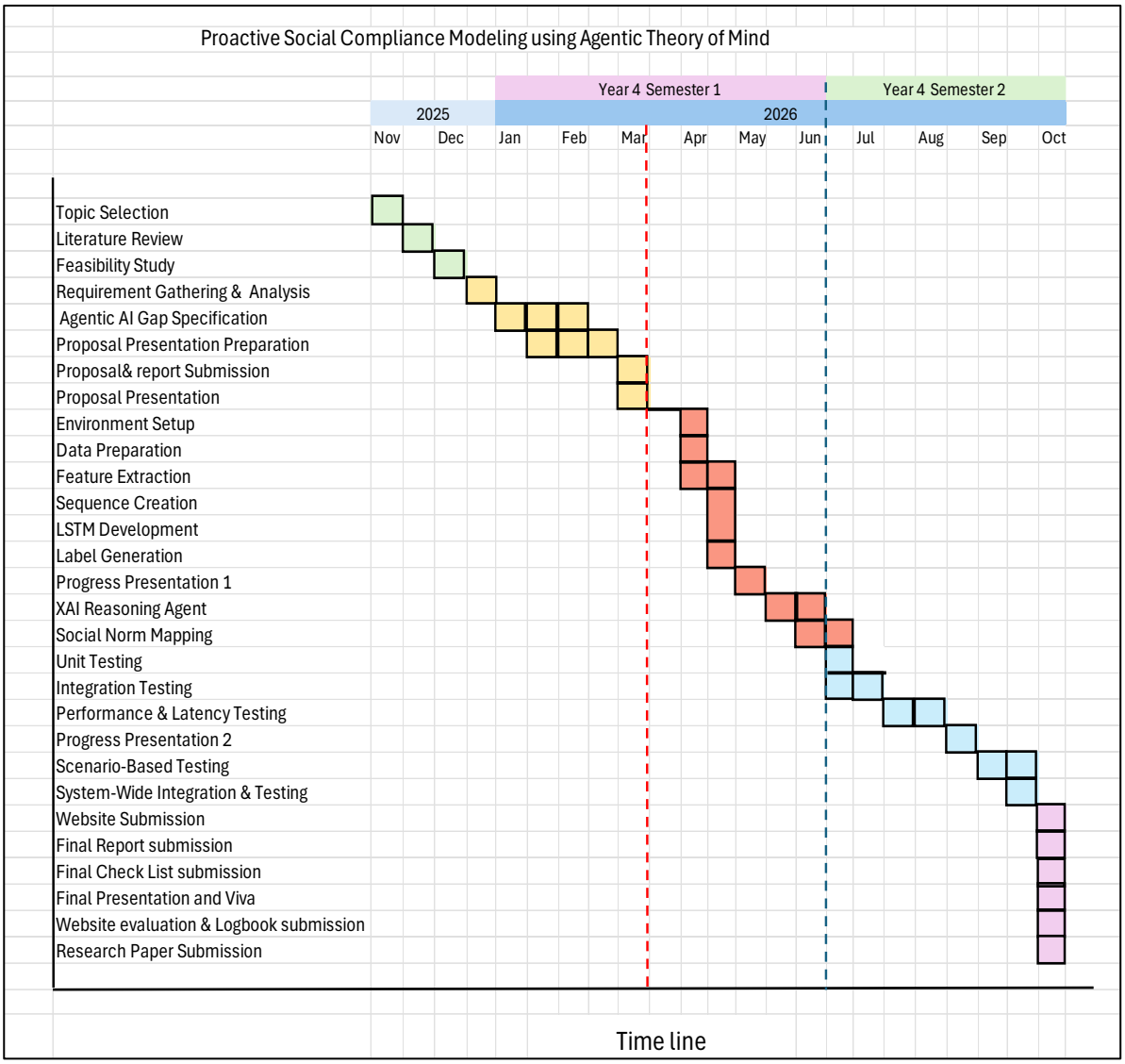


Figure 4:Gantt Chart

Figure 5:Gantt Chart

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